

Resources and readings

Here are some essential resources for data scientists. As a data scientist, you should be familiar with the following:

- Fundamentals of Data Visualization (<https://clauswilke.com/dataviz/>)
- Introduction to Numpy (<https://dujiaxin.github.io/gvsu-cis635-26w.github.io/samples/numpy.html>).
- Introduction to Pandas (<https://dujiaxin.github.io/gvsu-cis635-26w.github.io/samples/pandas.html>).
- Math Essentials
(https://courses.washington.edu/css490/2012.Winter/lecture_slides/02_math_essentials.pdf)
- Probability
(https://www.cs.princeton.edu/courses/archive/spring07/cos424/scribe_notes/0208.pdf).
- NumPy Cheat Sheet (<https://media.geeksforgeeks.org/wp-content/uploads/20240104182515/NumPy-Cheat-Sheet.pdf>).
- Pandas Cheat Sheet (https://pandas.pydata.org/Pandas_Cheat_Sheet.pdf).
- scikit-learn Cheat Sheet
(https://s3.amazonaws.com/assets.datacamp.com/blog_assets/Scikit_Learn_Cheat_Sheet_Python.pdf).
- matplotlib Cheat Sheet (<https://matplotlib.org/cheatsheets/>).

Why do we divide by $n - 1$ for Sample Variance?

When calculating population variance, we divide by N . But when working with a sample, we divide by $n - 1$ (Bessel's Correction). Why is that?

POPULATION VARIANCE (σ^2)

$$\sigma^2 = \sum (x_i - \mu)^2 / N$$

Uses the **True Mean (μ)** and the full population size (N). We have perfect information.

SAMPLE VARIANCE (s^2)

$$s^2 = \sum (x_i - \bar{x})^2 / (n - 1)$$

Uses the **Sample Mean ($\bar{x}$)**. We divide by $n-1$ to correct for bias.

1. The Visual Intuition: "The Centering Bias"

The core issue is that we don't know the True Mean (μ). We have to use the Sample Mean ($\bar{x}$) as a proxy. However, the Sample Mean is calculated specifically to be in the center of your sample data. This makes the data

specifically to be in the center of your sample data. This makes the data points look "closer" to the center than they actually are to the true population mean.

Try It Yourself

Click "Generate New Sample" to pick 5 random points from a hidden population (Mean $\mu = 50$).

Generate New Sample

True Population Mean (μ) 50.0

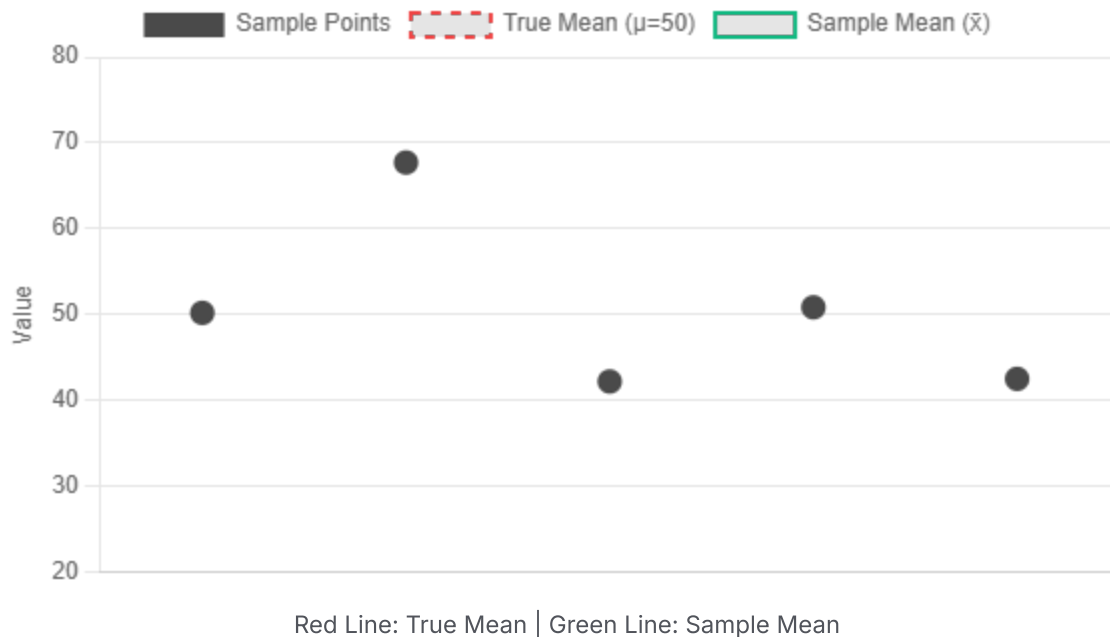
Your Sample Mean ($\bar{x}$) 50.7

SUM OF SQUARED ERRORS (DISTANCE)

From True Mean (μ): 430.7

From Sample Mean ($\bar{x}$): 428.1

Bias: Using sample mean "saves" 2.6 error units.



Notice: The sum of squared distances to the Sample Mean (Green) is **always** smaller than or equal to the True Mean (Red). This "underestimation" is why dividing by 'n' gives a result that is too small.

2. The Mathematical Proof

We can prove exactly how much we "lose" by using the sample mean. Walk through the derivation step-by-step.

Step 1: The Algebraic Expansion

We start with the sum of squares formula. Directly calculating the expectation is hard, so we use a standard algebraic identity to split it up.

Original Formula:

$$\sum (x_i - \bar{x})^2$$



$$\sum x_i^2 - n\bar{x}^2$$

We expanded the square and simplified. Now we have two separate terms to analyze.